CAPSTONE 8

AI PCB Design Review Checklist Assistant

Project report • Beginner-friendly review of three simple PCB designs

Project question
Can AI help beginners review a simple PCB design using a checklist, and what mistakes does it miss?

Prepared as a worked academic submission package
This report provides three representative design cases, a source-grounded checklist, sample AI-versus-human review analysis, prompt library, a one-page guide, and a scoring workbook. Since no actual PCB layout files or teacher comments were supplied, the design cases and review outcomes below are **illustrative desk-review examples**, not claims that physical boards were inspected or that a teacher performed a review. Replace the example review results with your own KiCad screenshots and teacher feedback before final submission.

Student: _____ Class/Roll No.: _____
Course: _____ Date: _____

1. Executive summary

This capstone explores a practical, low-risk way to use generative AI as a first-pass assistant for beginner PCB reviews. The assistant converts reliable layout guidance into explicit checks, asks for evidence, labels uncertainty, and directs the reviewer to electrical-rule checks (ERC), design-rule checks (DRC), calculations, datasheets, and human review. It should not declare a board safe or production-ready based only on a text description or screenshot.

Three small example designs are used: (A) a 5 V linear regulator board, (B) a current-limited LED indicator board, and (C) a 555 timer astable board. The checklist covers decoupling, power and return paths, trace width/current, clearances, component orientation/footprints, thermal considerations, connectors, and documentation.

Main finding: AI is useful for organizing checks and explaining likely issues, but it can miss hidden net connectivity, exact footprint dimensions, electrical limits, manufacturing constraints, and context-specific grounding decisions. False alarms can occur when it assumes a capacitor is missing from a cropped screenshot or treats a general rule as universal. The safest workflow is AI suggestions → source verification → KiCad ERC/DRC and calculations → human/teacher review.

Keywords: PCB layout, AI-assisted engineering, decoupling, grounding, trace width, DRC, verification, prompt engineering.

2. Scope, assumptions and method

Scope

- Beginner-level, low-voltage educational examples only; no claim of compliance or suitability for a real product.
- Review evidence consists of short design descriptions and expected review concerns. The report does not invent actual board screenshots or electrical measurements.
- Three designs are described textually so the checklist can be reused when screenshots or KiCad project files become available.
- Review categories are: critical/high, medium, low, false alarm, and needs evidence.

Workflow

Stage	Action	Evidence/output
1. Select cases	Choose regulator, LED indicator and 555 timer examples	Three short design briefs
2. Research	Read semiconductor vendor layout guidance, KiCad documentation and beginner review guides	Source-grounded review guidance
3. Draft checklist	Convert guidance into observable yes/no/unknown checks	Workbook checklist
4. AI first pass	Use a structured prompt and require evidence + confidence	Candidate findings
5. Verify	Check each claim against datasheet/source, schematic, PCB, ERC/DRC and calculations	Verified, rejected or unresolved
6. Compare	Compare AI suggestions with human/teacher review	Review table and error analysis
7. Automate	Use spreadsheet dropdowns and automatic score/status formulas	Scoring workbook
8. Report	Summarize limitations and safe workflow	Report and 7-slide presentation

Important boundary: without the actual KiCad board, netlist, design rules, stack-up, copper weight, operating current and component datasheets, a reviewer cannot validate exact clearances, trace widths, thermal performance, or connectivity.

3. Three selected example designs

Design A — 5 V linear regulator board

Purpose: convert a suitable DC input to regulated 5 V using a linear regulator (for example, a regulator in the 78xx family), input/output capacitors, input connector and output connector. **Review focus:** correct regulator pinout and package, capacitor values and placement per the exact datasheet, input/output polarity, copper area/thermal dissipation, current path width, and connector labels. A linear regulator can dissipate significant heat; check the power equation P ≈ (Vin – Vout) × Iout and the device's thermal limits. Do not assume every 5 V regulator has the

same pinout or capacitor requirements.

Design B — current-limited LED indicator

Purpose: power an LED from a low-voltage DC source through a series resistor. **Review focus:** resistor is actually in series, LED polarity matches the footprint/schematic, resistor value is calculated for supply voltage and LED forward voltage, resistor power rating is adequate, and connector polarity is clear. First-order estimate: R = (Vsupply – Vf) / ILED. Confirm the LED datasheet and select a conservative current; the formula is not a substitute for datasheet review.

Design C — 555 timer astable board

Purpose: create a repeating output using a 555 timer, timing resistors/capacitor, supply decoupling and an output connector or LED load. **Review focus:** correct 555 variant and pin mapping, short local bypass capacitor path, timing capacitor polarity if electrolytic, RESET/CONTROL pin treatment per datasheet, output load current, and keeping the timing node away from noisy/high-current routes where practical. Exact pin recommendations must be checked against the selected manufacturer and variant.

Design	Most important checks	Likely limitation of screenshot-only review
A. 5 V regulator	Pinout, capacitor placement/values, heat, current path	AI may not know regulator variant, current, copper area or thermal conditions
B. LED indicator	Series resistor, polarity, resistor power, connector labels	AI may misread silkscreen or fail to calculate current/power correctly
C. 555 timer	Pin mapping, decoupling, timing network, output load	AI may assume a generic 555 pinout or invent a universal grounding rule

4. Research-backed PCB review rules

Rule family	Verified practical rule	Evidence / how to verify
Decoupling	Place the local bypass capacitor close to the IC supply pin and keep its ground connection short; use the datasheet's recommended value and layout.	TI layout guidance [1]
Grounding/return	Ensure each signal has a sensible return path; avoid narrow bottlenecks, splits and planes that force return current around unintended loops.	Analog Devices explains ground loops and return paths [3]
Trace width/current	Choose width based on current, copper thickness, allowed temperature rise, and whether the trace is on an internal or external layer.	Use a recognised calculator, board constraints and manufacturer capability; Saturn PCB Toolkit [4]
Clearance/DRC	Set net classes and manufacturer-compatible clearance/width rules, then run DRC before treating the board as ready.	KiCad routing and DRC documentation [5] describes the relevant checks
Thermal	Estimate regulator/resistor power and check datasheet limits, copper area, heatsinking and ambient conditions.	Use the package thermal resistance and power calculations; AI alone cannot estimate these reliably
Footprints/connectors	Verify pad numbering, package dimensions, pin-1/polarity marks, connector orientation and mechanical fit against the datasheet or drawing.	Compare schematic symbol, footprint, 3D view and manufacturer drawing; screenshots can show obvious mismatches

Sources (accessed 25 October 2026):
[1] Texas Instruments, *CC13xx/CC26xx Hardware Configuration and PCB Design Considerations*: https://e2e.ti.com/cfs-file/__key/communityserver-discussions-components-files/538/26xx-layout-considerations.pdf
[2] Texas Instruments, *High-Speed Layout Guidelines*: https://www.ti.com/lit/an/scaa082a/scaa082a.pdf
[3] Analog Devices, *What Are the Basic Guidelines for Layout Design of Mixed-Signal PCBs?*: https://www.analog.com/en/resources/analog-dialogue/articles/what-are-the-basic-guidelines-for-layout-design-of-mixed-signal-pcbs.html
[4] Saturn PCB Toolkit: https://saturnpcb.com/saturn-pcb-toolkit/ (tool reference; validate against the actual design and current standards)
[5] KiCad PCB routing documentation: https://github.com/KiCad/kicad-doc/blob/master/src/pcbnew/pcbnew_routing.adoc

5. Verified checklist

ID	Check item	Status	Evidence / notes
C01	Input voltage range, output voltage/current and operating environment are documented.	Not checked	
C02	Exact component manufacturer/part numbers and datasheets are identified.	Not checked	
C03	ERC is run; power pins, net labels, connector pins and component values are checked.	Not checked	
C04	Footprints match the selected packages; pin numbering and polarity are verified.	Not checked	
C05	Required bypass capacitors are present and close to relevant supply pins.	Not checked	
C06	Capacitor ground return is short/direct; capacitor voltage rating and polarity are suitable.	Not checked	
C07	Track widths are justified by current, copper and temperature-rise assumptions.	Not checked	
C08	Clearances, via sizes, drill sizes and copper-to-edge constraints match manufacturer capabilities.	Not checked	

ID	Check item	Status	Evidence / notes
C09	High-current and sensitive/timing paths are routed with their circuit needs in mind	Not checked	
C10	Return paths are continuous and do not cross avoidable splits/slots or narrow bottlenecks	Not checked	
C11	Regulator and resistor dissipation are calculated and checked against datasheet limits	Not checked	
C12	Connector orientation, board outline, mounting holes and component clearance are checked	Not checked	
C13	Reference designators, polarity, pin 1, connector labels and warnings are readable	Not checked	
C14	KiCad DRC is run with the correct rules; every violation is fixed or documented with a justified waiver.	Not checked	
C15	Gerbers/drill outputs are inspected; BOM, schematic, board and revision are consistent	Not checked	
C16	A human reviewer checks high-risk findings and signs off before fabrication/use.	Not checked	

Status options for the workbook: Pass, Fail, Unknown, Not applicable. Unknown is not a pass. Any critical uncertainty should block sign-off until resolved.

6. Review results table — illustrative worked example

The following is a demonstration of how findings should be recorded, not a claim that physical PCB layouts or a teacher review were completed. Replace each example with the actual result after reviewing your own KiCad screenshots/projects.

Design	Candidate finding	Priority	Verification / decision	Outcome
A: Regulator	Input/output capacitors may be too far from regulator	High	Measure placement and compare with exact regulator datasheet guidance	Needs datasheet evidence and layout guidance
A: Regulator	Thermal dissipation may be excessive	High	Calculate (Vin–Vout)×Iout; check datasheet thermal limits and copper area	Needs calculation
A: Regulator	AI says all ground planes must be split	False alarm / invented rule	General rule is not universal; follow circuit-specific return guidance and [3]	Reject universal claim; use return-current analysis
B: LED	Series resistor may be missing or bypassed	High	Trace net connectivity in schematic/netlist; do not infer from cropped image alone	Needs schematic evidence
B: LED	LED polarity may be reversed	Medium	Check LED symbol, footprint pad numbers and physical polarity marking	Needs datasheet evidence
B: LED	AI recommends one fixed trace width for every board	Invented rule	Use current, copper, layer and temperature-rise inputs in [4]	Reject fixed trace-width rule
C: 555	Local supply bypass capacitor may be absent/far away	Medium	Check schematic and placement against selected 555 datasheet	Needs datasheet evidence
C: 555	RESET/CONTROL treatment is wrong	Medium	Verify selected device datasheet and schematic; do not assume generic	Needs datasheet evidence
All	DRC passes, so design is safe	False confidence	DRC checks only configured geometry/electrical constraints; requires manual review [5]	Reject; requires human review

7. AI tools and prompt-style comparison

Tools selected for the workflow: ChatGPT (structured review and report drafting), Google Gemini (independent second-opinion review), Perplexity (source discovery), and KiCad ERC/DRC (deterministic design-rule checks). The spreadsheet provides an additional lightweight automation step. The comparison below is a proposed evaluation protocol with illustrative expectations; it is not a measured benchmark of current model versions.

Tool	Best role	Strength to test	Risk / control
ChatGPT	Checklist application and structured findings	Can format issues as severity, evidence needed and recommended action	May hallucinate datasheet details; require URLs/part numbers and ve
Google Gemini	Independent second review	Can surface alternate concerns from the same evidence	May agree with prior assumptions or misread images; use same rubr
Perplexity	Find primary sources and trace claims	Useful for locating vendor documentation	Search snippets are not proof; open and verify the actual source
KiCad ERC/DRC	Deterministic circuit/layout checks	Reports violations against configured rules	Cannot assess every functional, thermal, EMI or mechanical issue

Prompt styles

Style	Example prompt	Expected result
Generic	“Review this PCB.”	Broad, inconsistent advice; weak traceability
Checklist-based	“Review only the listed items. For each, return Pass/Fail/Unknown, evidence, confidence and suitable.”	More consistent and auditable
Adversarial verification	“List assumptions, possible false alarms and what cannot be inferred from this image. Do not invent values.”	Better image limits; still requires external verification

For a real comparison, submit the same design evidence and same checklist to ChatGPT and Gemini, save both full responses, then score each finding: (1) correct and supported, (2) plausible but unverified, (3) false alarm, or (4) missed issue. Do not report an accuracy percentage until every finding has been independently adjudicated.

8. Error and miss analysis

Type	Example	Why it happens	Mitigation
Missed error	Wrong footprint pad mapping not noticed in screenshots	Pin numbers and hidden connectivity are difficult to inspect visually	Open footprint, schematic, netlist and datasheet; run ERC/DRC
Missed error	Trace overheats despite apparently neat routing	Current, copper thickness, layer and thermal environment are unknown	Calculate trace current/temperature rise and check power/therm
False alarm	AI claims a capacitor is missing because the component is not visible in the cropped image	Incomplete image/context	Provide full board, schematic and BOM; label crop boundaries
Invented rule	“Every PCB must have split analog/digital grounds”	Generalization from one design situation	Check authoritative device guidance and return-current behavio
False confidence	“DRC clean means production-ready”	Tool scope is mistaken for complete engineering validation	Use DRC + ERC + datasheets + calculations + manufacturing r
Incorrect assumption	AI assumes generic 555 or regulator pinout	Different vendors/packages/variants may differ	Require exact manufacturer part number and datasheet

Suggested measures for the actual experiment: verified true positives; false positives; missed issues; unsupported claims; and proportion of findings with valid source/evidence. Report counts separately for each design and each AI tool. With only three small examples, describe results as a classroom pilot rather than a general performance claim.

9. Final one-page guide: safe AI-assisted PCB review

Before asking AI

- Collect the complete schematic, PCB top/bottom screenshots, board outline, BOM, exact part numbers, intended input/output, maximum current, copper weight/layers and manufacturer rules.
- Remove private project information if not needed. Use readable, full-board and close-up views; state what is not visible.
- Set the expected manufacturer limits in KiCad before running DRC.

Ask AI to

- Apply a fixed checklist and report Pass / Fail / Unknown / N/A.
- Separate observations from assumptions; give the evidence location and confidence for every concern.
- Identify which checks require datasheets, calculations, netlist inspection or physical measurement.
- List potential false alarms and avoid universal claims unless supported by a cited source.

Verify independently

- Open cited vendor datasheets and confirm the exact part number and page/section.
- Run ERC and DRC in KiCad; inspect every violation and any waiver.
- Calculate resistor power, regulator dissipation and trace-current needs using documented assumptions.
- Inspect footprints, polarity, connector orientation, board edges, silkscreen and manufacturing outputs.
- Ask a teacher/experienced reviewer to check unresolved high-risk items.

Decision rule: AI can help find questions to investigate; it cannot certify the board. Never treat an AI “Pass” or a clean DRC report as proof of complete safety or functionality.

10. Automation: spreadsheet scoring

The companion workbook automates a simple review completeness score. For each checklist row, select Pass, Fail, Unknown or N/A. Score = Pass items ÷ all applicable checked items × 100. Unknown and Fail remain unresolved; they do not count as passes. The workbook also displays a status summary. This score is a process-completeness indicator only—not a measure of electrical safety or design quality.

Suggested workflow: (1) fill design metadata, (2) enter evidence in the notes column, (3) choose a status for each check, (4) filter Fail/Unknown, (5) attach supporting screenshots or datasheet references, (6) obtain human sign-off. Before submitting, confirm formulas work in your spreadsheet application and include a screenshot of the populated scoring sheet.

11. Before/after analysis

Before AI/checklist	After structured AI + verification
Review is informal and may overlook documentation, grounding or thermal checks.	A repeatable checklist ensures the same review categories are considered each time.
Advice may be vague (“make traces wider”).	The workflow asks for current, copper, layer, temperature-rise and voltage-drop assumptions.
AI may state rules confidently without evidence.	Every rule must have a source, datasheet reference or explicit “needs verification” label.
A clean DRC may be interpreted as full approval.	DRC is one layer of verification alongside ERC, calculations, datasheets and human review.
Review results are hard to compare across tools.	A shared rubric records verified findings, false alarms, missed issues and unsupported claims.

12. Prompt library

Prompt 1 — checklist creation

Act as a cautious PCB design-review assistant for beginners. Create a checklist for low-voltage educational PCB layouts covering decoupling, return paths/grounding, trace width/current, clearances, footprint/polarity, thermal limits, connectors, silkscreen, ERC/DRC and manufacturing outputs. For each item give: why it matters, evidence required, and a source type to consult. Do not give universal numeric rules without design inputs.

Prompt 2 — design review

Review the attached schematic/PCB evidence using only the provided checklist. For every item return: ID, status (Pass/Fail/Unknown/N/A), observation, evidence location, severity, confidence, verification step and source. If an item cannot be determined from the image, mark Unknown. Do not invent component values, pin numbers, net connectivity, clearances or trace widths.

Prompt 3 — source verification

For each technical claim, identify the exact primary source and the relevant section/page. Prefer the selected component manufacturer datasheet, vendor layout guide, KiCad documentation or a recognized calculation method. If the source cannot be verified, label the claim unsupported and remove it from the verified checklist.

Prompt 4 — false-alarm challenge

Audit your previous review. List possible false positives, missed issues, hidden assumptions and any universal rule that may depend on context. For each item explain what additional file, measurement or datasheet would resolve it. Do not defend an earlier claim merely because you produced it.

Prompt 5 — second-model comparison

Review the same evidence independently. Do not see the first model's findings. Use the same checklist and output format. After both reviews are complete, compare the findings against an independent adjudication table. Do not call an issue correct unless evidence supports it.

Prompt 6 — final report

Summarize verified findings, unresolved items, false alarms, missed issues and tool limitations. Separate actual observed evidence from hypothetical examples. Include a short safety statement that AI suggestions do not replace ERC/DRC, datasheet checks, calculations or expert review.

13. Limitations and conclusion

AI cannot reliably infer hidden copper layers, exact net connectivity, footprint pad numbering, manufacturing tolerances, component thermal limits or electrical behavior from a cropped image alone. It may hallucinate sources or part specifications, overgeneralize grounding advice, miss subtle visual errors, or flag a feature that is simply outside

the screenshot. Image quality, context, model version and prompt wording all affect the result.

The recommended division of responsibility is: AI for checklist organization and candidate observations; vendor datasheets and standards for technical requirements; KiCad ERC/DRC for configured rule checks; calculations and simulation for quantitative questions; and a teacher/qualified engineer for final judgment. This is an educational aid, not a design certification.

Conclusion: AI can make beginner PCB reviews more systematic when the task is constrained by a source-grounded checklist and every finding is verified. Its greatest value is helping users ask better questions and document the evidence. Its biggest risk is confident but unsupported advice. Human review and electronic design verification remain essential.

Appendix A — human/teacher review record

Field	To complete during actual review
Reviewer / role	_____
Design and revision	_____
Files reviewed	Schematic / PCB / BOM / datasheets / Gerbers (circle or list)
High-priority findings	_____
AI findings rejected as false alarms	_____
Issues AI missed	_____
Unresolved questions	_____
Reviewer comments / date	_____

Appendix B — final submission checklist

- Verified checklist workbook completed with evidence and status for each item.
- Review results table updated with actual ChatGPT and Gemini outputs.
- Error/miss analysis supported by the teacher or independent reviewer’s notes.
- One-page guide included in this report.
- Presentation updated with screenshots from the three actual designs.
- Automation workbook tested and screenshot added.
- Prompt library retained with full prompts and tool/version/date details.
- Verification report includes source links, ERC/DRC evidence and calculations.
- Before/after analysis uses actual initial and final review artifacts.
- All illustrative examples are clearly replaced or labelled before submission.